

ProbOS — Month 7 Pre-Plan

Pharma GPU and C++ Parity

Nisong Monyimba

July 8, 2026

Month goal

Bring `PharmaStabilityModel` to the same GPU and C++ performance-path maturity as `BatteryModel2Cell`.

Week 26 – GPU dispatch for `PharmaStabilityModel`

- CuPy dispatch mirroring the battery model’s `xp` pattern exactly
- Honest benchmark against CPU/C++ – do NOT assume GPU wins

Week 27 – C++ port

- `PharmaCell`-equivalent in C++
- Native GoogleTest coverage from day one
- `pybind11` bindings

Week 28 – Cross-language validation

- Statistical agreement, not bit-exact, given different RNGs

Week 29 – Dashboard integration

Study materials for this month: Primarily applying already-demonstrated skills from Months 1–3 – minimal new study material needed.

- Optional: Vandevor, Josuttis, Gregor, *C++ Templates: The Complete Guide* (2nd ed.) – if pursuing the `Normal<T>` autodiff future direction.

Exit criteria

Pharma has full GPU and C++ parity with battery, honestly benchmarked.